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Domain Shift and Generalization Strategies for Foundation Models in Clinical Imaging

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Abstract

The advent of foundation models—large-scale, pretrained deep learning systems—has transformed the landscape of machine learning by enabling general-purpose representations that can be efficiently adapted to a wide range of downstream tasks. In the context of medical imaging, these models hold immense promise for addressing longstanding challenges in scalability, generalization, and data scarcity. However, deploying foundation models across the diverse and heterogenous landscape of clinical environments necessitates robust strategies for domain adaptation and generalization. Domain shifts, stemming from variations in patient demographics, imaging devices, acquisition protocols, and disease prevalence, often result in significant degradation in model performance when applied out-of-distribution. To bridge this gap, a variety of adaptation techniques—including fine-tuning, prompt engineering, adapter-based learning, test-time training, and semisupervised transfer—have been developed and tailored for high-stakes medical applications. This survey provides a comprehensive overview of domain adaptation and generalization strategies in the era of foundation models, with a specific focus on their application to medical imaging. We begin by framing the problem of domain shift in clinical settings and reviewing classical theories of generalization and adaptation. We then explore how these ideas extend—or break down—when applied to large pretrained models such as vision transformers (ViTs), contrastive encoders, and multimodal architectures trained on massive medical and non-medical corpora. Next, we examine the practical methods used to adapt foundation models to specific clinical domains, analyzing their efficiency, scalability, and empirical performance across a wide range of imaging modalities, including radiography, CT, MRI, ultrasound, and pathology. We also highlight emerging evaluation benchmarks and discuss the limitations of current adaptation paradigms, particularly in low-resource and federated environments. Beyond technical considerations, we delve into the ethical, regulatory, and operational implications of adapting foundation models in healthcare. These include issues related to model interpretability, privacy-preserving adaptation, certification of evolving models, and bias mitigation across underrepresented populations. Finally, we identify key open research directions-including theoretical advancements in deep domain generalization, efficient continual adaptation, multi-modal integration, and human-in-the-loop learning-and propose a forward-looking vision for safe, robust, and equitable AI in medical imaging. By synthesizing a rapidly growing body of work, this survey aims to guide researchers, clinicians, and developers in understanding the foundational and translational challenges of domain adaptation in the age of foundation models.

Keywords

clinical ai, domain adaptation, domain generalization, few-shot learning, foundation models, general topics for engineers, medical imaging, model robustness, multimodal ai, out-of-distribution generalization, representation learning, self-supervised learning, transfer learning, vision transformers

Domain Shift and Generalization Strategies for Foundation Models in Clinical Imaging

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Abstract

The advent of foundation models—large-scale, pretrained deep learning systems—has transformed the landscape of machine learning by enabling general-purpose representations that can be efficiently adapted to a wide range of downstream tasks. In the context of medical imaging, these models hold immense promise for addressing longstanding challenges in scalability, generalization, and data scarcity. However, deploying foundation models across the diverse and heterogenous landscape of clinical environments necessitates robust strategies for domain adaptation and generalization. Domain shifts, stemming from variations in patient demographics, imaging devices, acquisition protocols, and disease prevalence, often result in significant degradation in model performance when applied out-of-distribution. To bridge this gap, a variety of adaptation techniques—including fine-tuning, prompt engineering, adapter-based learning, test-time training, and semi-supervised transfer—have been developed and tailored for high-stakes medical applications. This survey provides a comprehensive overview of domain adaptation and generalization strategies in the era of foundation models, with a specific focus on their application to medical imaging. We begin by framing the problem of domain shift in clinical settings and reviewing classical theories of generalization and adaptation. We then explore how these ideas extend—or break down—when applied to large pretrained models such as vision transformers (ViTs), contrastive encoders, and multimodal architectures trained on massive medical and non-medical corpora. Next, we examine the practical methods used to adapt foundation models to specific clinical domains, analyzing their efficiency, scalability, and empirical performance across a wide range of imaging modalities, including radiography, CT, MRI, ultrasound, and pathology. We also highlight emerging evaluation benchmarks and discuss the limitations of current adaptation paradigms, particularly in low-resource and federated environments. Beyond technical considerations, we delve into the ethical, regulatory, and operational implications of adapting foundation models in healthcare. These include issues related to model interpretability, privacy-preserving adaptation, certification of evolving models, and bias mitigation across underrepresented populations. Finally, we identify key open research directions—including theoretical advancements in deep domain generalization, efficient continual adaptation, multi-modal integration, and human-in-the-loop learning—and propose a forward-looking vision for safe, robust, and equitable AI in medical imaging. By synthesizing a rapidly growing body of work, this survey aims to guide researchers, clinicians, and developers in understanding the foundational and translational challenges of domain adaptation in the age of foundation models.

Keywords: Foundation Models, Medical Imaging, Domain Adaptation, Domain Generalization, Transfer Learning, Vision Transformers, Self-supervised Learning, Multimodal AI, Clinical AI, Representation Learning, Few-shot Learning, Model Robustness, Out-of-Distribution Generalization

1 Introduction

Medical imaging plays a crucial role in modern healthcare, serving as a foundational tool for diagnosis, prognosis, treatment planning, and disease monitoring. Over the past decade, the rapid advancement of machine learning, particularly deep learning, has significantly transformed the medical imaging landscape, enabling automated analysis of complex imaging data with unprecedented accuracy. Convolutional Neural Networks (CNNs) and, more recently, Transformer-based architectures have demonstrated remarkable success across a variety of medical imaging modalities, including Magnetic Resonance Imaging (MRI), Computed Tomography (CT), Ultrasound, Positron Emission Tomography (PET), and X-ray [1]. Despite these advances, a persistent and critical challenge remains: ensuring robust model performance across diverse clinical settings, patient populations, imaging devices, and acquisition protocols [2]. This challenge is often referred to in the literature as the problem of *domain shift*, and its mitigation is addressed through the twin paradigms of *domain adaptation* and *domain generalization*. Domain adaptation (DA) and domain generalization (DG) aim to develop models that maintain high performance when deployed on data that is statistically different from the training distribution. In medical imaging, domain shifts may arise due to factors such as scanner manufacturers, acquisition parameters, anatomical and pathological variations across populations, site-specific workflows, and even regional differences in healthcare infrastructure. A model trained on a dataset from one institution often exhibits degraded performance when applied to data from another, which severely hinders the translation of artificial intelligence (AI) models into real-world clinical practice. Consequently, domain robustness is not merely a technical desideratum—it is a prerequisite for clinical applicability, patient safety, and regulatory approval [3]. Recently, the advent of *foundation models* has introduced a paradigm shift in machine learning. These are large-scale, pre-trained models that can be adapted to a wide range of downstream tasks with minimal fine-tuning [4]. Foundation models, particularly those based on Transformer architectures, have achieved state-of-the-art performance in natural language processing (e.g., BERT, GPT) and computer vision (e.g., CLIP, DINO, SAM) [5]. In the medical imaging domain, foundation models such as BioViL, MedCLIP, and MedSAM have shown promising results by leveraging large-scale multi-modal data and self-supervised pretraining to capture generalized visual representations. These models offer a potential pathway to overcoming the domain shift problem by providing robust, transferable features that are less sensitive to variations in the data distribution [6]. The intersection of domain adaptation/generalization and foundation models in medical imaging is thus a highly promising and rapidly evolving area of research [7]. Unlike conventional domain adaptation approaches that require access to target domain data for fine-tuning or adversarial alignment, foundation models aim to generalize across domains inherently through extensive pretraining on diverse and heterogeneous datasets. This shifts the focus from model-specific solutions to data-centric and representation-centric strategies, aligning with the broader trend in AI toward more scalable and adaptable systems. Despite their promise, several questions remain unanswered. What are the mechanisms by which foundation models enable domain generalization? How effective are they in truly out-of-distribution scenarios, particularly in safety-critical medical applications? What are the trade-offs between task-specific adaptation and general-purpose representation learning [8]? How do we evaluate generalization in a meaningful way, especially when ground truth annotations are scarce or inconsistent across domains? Furthermore, how can we integrate domain knowledge, such as anatomical priors or clinical guidelines, into these large models to enhance both their interpretability and performance? This survey aims to provide a comprehensive and critical overview of domain adaptation and generalization in medical imaging with a specific focus on the emerging role of foundation models. We begin by reviewing the fundamental concepts of domain shift and its implications in medical imaging. We then discuss traditional approaches to domain adaptation and generalization, including supervised, unsupervised, and semi-supervised methods, as well as strategies based on data augmentation, feature alignment, and meta-learning. Subsequently, we delve into the architecture, training paradigms, and deployment considerations of foundation models in medical imaging [9]. Particular attention is paid to self-supervised and multi-modal learning frameworks, as these are most closely aligned with the goals of cross-domain robustness. We also provide a detailed taxonomy of recent works at the intersection of foundation models and domain robustness, categorizing them based on their methodological contributions, target tasks (e.g., classification, segmentation, detection), imaging modalities, and evaluation strategies [10]. Through this taxonomy, we identify emerging patterns, current limitations, and under-explored research directions [11]. Additionally, we discuss ethical, regulatory, and clinical translation challenges that arise when deploying large, opaque models in real-world healthcare environments. In summary, this survey seeks to elucidate the state of the art and future directions in developing domain-adaptive and generalizable models for medical imaging, with an emphasis on leveraging the power of foundation models. By integrating insights from machine learning, computer vision, and medical imaging communities, we aim to provide a valuable resource for researchers and practitioners seeking to build robust, scalable, and clinically meaningful AI systems [12].

2 Background and Problem Formulation

The tasks of domain adaptation (DA) and domain generalization (DG) can be rigorously framed within the formalism of statistical learning theory. Let $\mathcal{X}$ denote the input space (e.g., medical images) and $\mathcal{Y}$ the output space (e.g., diagnostic labels, segmentation masks) [13]. A domain $\mathcal{D}$ is defined as a joint distribution $\mathcal{D} = P(X, Y)$ over $\mathcal{X} \times \mathcal{Y}$ [14]. In supervised learning, we assume access to a labeled dataset $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$ sampled i.i.d. from a source domain $\mathcal{D}_S$. The learner aims to train a predictive function $f_\theta : \mathcal{X} \to \mathcal{Y}$, parameterized by θ, such that the expected risk $\mathcal{R}_{\mathcal{D}}(f_\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(f_\theta(x), y)]$ is minimized, where $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$ is a task-specific loss function [15]. In the domain adaptation setting, the primary challenge arises from the fact that the training and test distributions are not identical; that is, we have a labeled source domain $\mathcal{D}_S$ and an unlabeled or sparsely labeled target domain $\mathcal{D}_T$ such that $\mathcal{D}_S \neq \mathcal{D}_T$. The goal is to minimize the target risk $\mathcal{R}_{\mathcal{D}_T}(f_\theta)$ even though the model is primarily trained on $\mathcal{D}_S$ [16]. The discrepancy between the marginal distributions $P_S(X)$ and $P_T(X)$, as well as the conditional distributions $P_S(Y|X)$ and $P_T(Y|X)$, necessitates the development of strategies that can bridge the domain gap. Traditional approaches include instance reweighting, feature space alignment, adversarial training, and domain-invariant representation learning. Domain generalization, in contrast, assumes access to multiple labeled source domains $\{\mathcal{D}_S^{(i)}\}_{i=1}^N$ and no access to the target domain during training. The aim is to learn a model f_θ that generalizes well to any unseen domain $\mathcal{D}_T$ drawn from a distribution over domains $\mathcal{P}_\mathcal{D}$. Formally, the objective becomes minimizing the expected target risk over the distribution of domains: $\mathbb{E}_{\mathcal{D}_T \sim \mathcal{P}_\mathcal{D}}[\mathcal{R}_{\mathcal{D}_T}(f_\theta)]$. This necessitates learning representations that are invariant not only to observed domain variations but also to unobserved factors of variation likely to be encountered in deployment scenarios. This is particularly crucial in medical imaging, where models must operate reliably across hospitals, scanner types, patient demographics, and imaging protocols, often with minimal opportunities for recalibration. In medical imaging, domain shifts can be subtle yet highly consequential. For example, a CT scanner from manufacturer A may produce images with slightly different contrast levels or spatial resolutions than a scanner from manufacturer B. Similarly, annotation styles can vary among radiologists or across institutions, introducing label noise and semantic inconsistencies [17, 18]. Let $\mathcal{X}_S$ and $\mathcal{X}_T$ denote the feature spaces of source and target domains, respectively [19]. Even if $\mathcal{X}_S \approx \mathcal{X}_T$ in terms of appearance, the underlying data-generating mechanisms may differ significantly, i.e., $P_S(X, Y) \neq P_T(X, Y)$ [20]. When this gap is not accounted for, a model trained to minimize $\mathcal{R}_{\mathcal{D}_S}(f_\theta)$ may yield high error on $\mathcal{D}_T$, potentially leading to harmful clinical outcomes. The theory of domain adaptation provides upper bounds on the target risk in terms of the source risk and a divergence term between domains. One well-known result is derived from the work of Ben-David et al., which states that for any hypothesis $f \in \mathcal{H}$, the target risk $\mathcal{R}_{\mathcal{D}_T}(f)$ can be bounded by:

$$\mathcal{R}_{\mathcal{D}_T}(f) \leq \mathcal{R}_{\mathcal{D}_S}(f) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) + \lambda,$$

where $d_{\mathcal{H}\Delta\mathcal{H}}$ is the $\mathcal{H}\Delta\mathcal{H}$-divergence between source and target distributions, and λ is the shared error of the ideal joint hypothesis on both domains [21]. This bound suggests that minimizing source risk alone is insufficient; the domain discrepancy must also be addressed, which motivates techniques such as domain adversarial neural networks (DANN) and maximum mean discrepancy (MMD) based methods. The arrival of foundation models changes the landscape of domain adaptation and generalization in fundamental ways. Foundation models are trained on large-scale datasets, often using self-supervised or weakly supervised objectives, with the goal of learning semantically rich and transferable representations. Let f_{θ_0} denote a foundation model pretrained on a large corpus $\mathcal{D}_{\text{pretrain}}$. The downstream adaptation process then involves fine-tuning f_{θ_0} to a target task using a smaller labeled dataset $\mathcal{D}_{\text{task}}$ [22]. The hypothesis is that the parameters θ_0 encode generalizable features that mitigate the impact of domain shift, i.e.,

$$\mathcal{R}_{\mathcal{D}_T}(f_{\theta_0}) - \mathcal{R}_{\mathcal{D}_S}(f_{\theta_0}) \ll \mathcal{R}_{\mathcal{D}_T}(f_\theta) - \mathcal{R}_{\mathcal{D}_S}(f_\theta),$$

for θ trained from scratch on $\mathcal{D}_S$ [23]. Moreover, foundation models often incorporate multi-modal information, such as paired image-text data, which can enrich the learned representation space and provide auxiliary semantic grounding. In medical imaging, this might include learning from radiology reports, anatomical atlases, or electronic health records, thereby anchoring visual features in clinical context and enhancing their robustness across domains. In conclusion, the formalization of domain adaptation and generalization provides a rigorous theoretical foundation for understanding the limitations of current models and the potential of foundation models. As we shall explore in the subsequent sections, incorporating inductive biases, leveraging self-supervised pretraining, and designing principled evaluation protocols are key to building generalizable AI systems for medical imaging.

3 Classical Approaches to Domain Adaptation and Generalization in Medical Imaging

Before the recent surge of foundation models, the research community devoted considerable effort to developing classical methods for addressing domain shift in medical imaging [24]. These techniques can be broadly categorized into four groups: input-level adaptation, feature-level alignment, output-level correction, and meta-learning-based generalization [25]. Each of these paradigms attempts to mitigate domain discrepancies through explicit or implicit modeling of inter-domain variability. While effective in specific contexts, these classical approaches often rely on carefully curated data and manual engineering, and their generalizability remains limited compared to emerging foundation model frameworks.

Input-Level Adaptation

Input-level adaptation methods aim to reduce domain shift by transforming the source domain data to mimic the target domain distribution in the image space. This is often achieved using image-to-image translation techniques such as CycleGAN or UNIT. Let $x_S \sim P_S(X)$ and $x_T \sim P_T(X)$ be source and target domain images, respectively. The goal is to learn a mapping $G : \mathcal{X}_S \to \mathcal{X}_T$ such that the translated images $G(x_S)$ are indistinguishable from x_T according to some discriminator D. Formally, the optimization problem becomes:

$$\min_G \max_D \mathbb{E}_{x_T}[\log D(x_T)] + \mathbb{E}_{x_S}[\log(1 - D(G(x_S)))] + \lambda \mathcal{L}_{\text{cycle}}(G),$$

where $\mathcal{L}_{\text{cycle}}$ enforces cycle-consistency to preserve anatomical and pathological content [26]. While such approaches have been successfully applied in cross-modality scenarios (e.g., CT to MRI), they are often sensitive to artifacts, hallucinations, and mode collapse, making clinical validation challenging.

Feature-Level Alignment

Feature-level alignment is one of the most popular strategies for domain adaptation in medical imaging. The underlying assumption is that a shared representation space exists in which features from both source and target domains are indistinguishable. Let $\phi : \mathcal{X} \to \mathbb{R}^d$ denote a feature extractor. The aim is to align the distributions $P_S(\phi(x))$ and $P_T(\phi(x))$ using statistical or adversarial objectives. A common statistical approach is based on minimizing Maximum Mean Discrepancy (MMD):

$$\mathcal{L}_{\text{MMD}} = \|\mathbb{E}_{x_S \sim \mathcal{D}_S}[\phi(x_S)] - \mathbb{E}_{x_T \sim \mathcal{D}_T}[\phi(x_T)]\|^2_{\mathcal{H}},$$

where $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS). Alternatively, adversarial alignment employs a domain discriminator D that attempts to distinguish features from different domains, while the feature extractor ϕ is trained to fool D:

$$\min_{\phi} \max_{D} \mathbb{E}_{x_S}[\log D(\phi(x_S))] + \mathbb{E}_{x_T}[\log(1 - D(\phi(x_T)))].$$

Such frameworks, including Domain-Adversarial Neural Networks (DANN) and ADDA, have shown promise in tasks like lesion segmentation and disease classification [27]. However, they often require access to target domain data during training and are sensitive to the choice of alignment loss and batch composition [28].

Output-Level and Hypothesis Alignment

Another classical approach involves aligning the output space across domains, particularly when feature distributions are hard to match directly. This is useful when source and target domain images differ significantly in appearance, but the semantic output space (e.g., organ segmentation masks or disease labels) remains structurally similar [29]. One strategy is entropy minimization on target domain predictions [30]. Given an unlabeled sample x_T, the model's output $f_\theta(x_T)$ should be confident (i.e., low entropy):

$$\mathcal{L}_{\text{ent}} = -\mathbb{E}_{x_T \sim \mathcal{D}_T} \sum_{k=1}^{K} f_\theta(x_T)_k \log f_\theta(x_T)_k,$$

where K is the number of classes. Another technique is self-training, where pseudo-labels are generated for the target domain using a confident source-trained model, and then the model is fine-tuned using these pseudo-labels. Such methods are especially useful in medical imaging, where acquiring pixel-wise annotations for tasks like segmentation is costly and labor-intensive.

Meta-Learning and Episodic Training

Meta-learning-based domain generalization focuses on learning models that can generalize to unseen domains by simulating domain shift during training [31]. In this framework, source domains are split into multiple pseudo-domains, and training is performed in an episodic manner to mimic test-time variability. Let $\{\mathcal{D}_S^{(i)}\}_{i=1}^N$ be source domains. During each episode, one domain is treated as a pseudo-target while the rest serve as pseudo-sources. The model is trained to optimize for performance on the held-out domain after being updated on the others. This approach is inspired by Model-Agnostic Meta-Learning (MAML) and can be expressed as:

$$\min_\theta \sum_{i=1}^N \mathcal{R}_{\mathcal{D}_S^{(i)}}(f_{\theta-\alpha\nabla_\theta \mathcal{R}_{\mathcal{D}_{-i}}(f_\theta)}),$$

where $\mathcal{D}_{-i}$ denotes all domains except i, and α is a learning rate [32]. This technique promotes learning domain-agnostic representations and has been successfully applied in robust segmentation of anatomical structures across multi-site datasets [33].

Limitations of Classical Methods

Despite their effectiveness in controlled settings, classical domain adaptation and generalization methods suffer from several limitations. Most notably, their success often hinges on the availability of representative target data or well-annotated multi-domain datasets, which are scarce in real-world medical imaging. Moreover, these methods typically involve complex training protocols, extensive hyperparameter tuning, and are brittle when faced with domain shifts that were not simulated during training. Additionally, many approaches assume a shared label space and semantic structure across domains, which may not hold in heterogeneous clinical datasets where taxonomies, annotation guidelines, and disease prevalence differ significantly [34]. As a result, there is growing consensus that scalable, representation-centric strategies—such as those enabled by foundation models—are required to address the full spectrum of domain shift in clinical AI systems. In the following section, we explore how foundation models offer a new paradigm that inherently incorporates the principles of data diversity, representation learning, and cross-domain robustness—potentially overcoming many of the limitations outlined above.

4 Foundation Models for Domain Adaptation and Generalization in Medical Imaging

Foundation models have emerged as a transformative paradigm in artificial intelligence, redefining how models are trained, adapted, and generalized across a wide range of tasks. In the context of medical imaging, foundation models hold the promise of overcoming longstanding limitations in domain adaptation and generalization by leveraging scale, diversity, and pretraining strategies rooted in self-supervision or multi-modal learning. These models are typically trained on massive and heterogeneous datasets spanning multiple domains, imaging modalities, and clinical tasks. Once pretrained, they can be adapted to downstream tasks with minimal supervision, often outperforming task-specific models that were explicitly designed for individual domains.

Architectural Principles and Pretraining Objectives

Foundation models in medical imaging are characterized by their large capacity, modular architecture, and general-purpose pretraining objectives [35]. Most commonly, these models adopt either transformer-based architectures (e.g., Vision Transformers, or ViTs) or hybrid CNN-transformer frameworks that combine local feature extraction with global attention mechanisms [36]. Let $\mathcal{F}_\theta : \mathcal{X} \to \mathbb{R}^d$ be a foundation encoder parameterized by θ, mapping input images $x \in \mathcal{X}$ to high-dimensional feature vectors [37]. Pretraining is often conducted using self-supervised learning (SSL) objectives [38]. One of the most widely used frameworks is contrastive learning, where the model learns to bring positive pairs (e.g., different augmentations of the same image) closer in feature space while pushing apart negative pairs. Formally, given two augmented views x_i and x_j of an image x, the contrastive loss can be written as:

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp(\text{sim}(\mathcal{F}_\theta(x_i), \mathcal{F}_\theta(x_j))/\tau)}{\sum_{k=1}^{N} \exp(\text{sim}(\mathcal{F}_\theta(x_i), \mathcal{F}_\theta(x_k))/\tau)},$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity function and τ is a temperature parameter. Other SSL strategies include masked image modeling (MIM), where random patches of an input image are masked and the model is trained to reconstruct the missing content, often using decoder heads or inpainting objectives. These methods have demonstrated strong performance in learning generalizable representations without requiring manual annotations.

Multi-Modal and Cross-Modal Pretraining

Foundation models also benefit significantly from multi-modal learning, where images are paired with auxiliary information such as radiology reports, clinical notes, or structured metadata. One prominent example is the use of vision-language pretraining (VLP), where models are trained to align image and text representations in a shared embedding space [39]. Let $\mathcal{F}_\theta^{\text{img}}$ and $\mathcal{F}_\phi^{\text{text}}$ denote image and text encoders, respectively [40]. The model is trained to maximize the similarity between paired image-text embeddings $(\mathcal{F}_\theta^{\text{img}}(x), \mathcal{F}_\phi^{\text{text}}(t))$ using objectives such as InfoNCE or contrastive language-image pretraining (CLIP):

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(\text{sim}(\mathcal{F}^{\text{img}}(x_i), \mathcal{F}^{\text{text}}(t_i))/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(\mathcal{F}^{\text{img}}(x_i), \mathcal{F}^{\text{text}}(t_j))/\tau)}.$$

By grounding visual features in natural language, these models are able to internalize semantic concepts, clinical terminology, and contextual knowledge that aid generalization to novel tasks and domains [41]. This is particularly valuable in zero-shot and few-shot scenarios, where task-specific supervision is unavailable.

Transfer Paradigms: Zero-Shot, Few-Shot, and Fine-Tuning

Once pretrained, foundation models can be adapted to new domains using various transfer learning paradigms. In the zero-shot setting, the model is directly applied to a target domain or task without any further training, relying on the robustness and universality of its learned representations. For example, in multi-label disease classification, the pretrained model can score each disease concept independently and return predictions without explicit task-specific retraining [42]. In few-shot transfer, the model is fine-tuned on a small labeled dataset $\mathcal{D}_{\text{few}} = \{(x_i, y_i)\}_{i=1}^{k}$ sampled from the target domain. Owing to the expressiveness of the pretrained encoder $\mathcal{F}_{\theta_0}$, only a lightweight adaptation head—such as a linear classifier or low-rank adapter—is required [43]. The training objective becomes:

$$\min_{\phi} \sum_{(x,y)\in\mathcal{D}_{\text{few}}} \ell(h_\phi(\mathcal{F}_{\theta_0}(x)), y),$$

where h_ϕ denotes the adaptation head and ℓ is a task-specific loss (e.g., cross-entropy or Dice loss). Full fine-tuning involves jointly optimizing the encoder and task-specific components using the target domain data. This approach is more computationally intensive but often yields superior performance when sufficient labeled data is available. Recent studies have also explored parameter-efficient fine-tuning using techniques such as adapters, prompt tuning, or LoRA (Low-Rank Adaptation), which reduce the number of trainable parameters and facilitate cross-domain portability [44].

Inductive Bias and Representation Robustness

A key strength of foundation models is their ability to learn transferable representations that capture the inductive biases necessary for generalization across diverse domains. These include invariance to image contrast, resolution, scanner manufacturer, patient demographics, and anatomical variation. Mathematically, let $z = \mathcal{F}_\theta(x)$ be the learned embedding of image x, and let $\delta(x)$ denote domain-specific perturbations (e.g., style, noise) [45]. Robustness implies that:

$$\|\mathcal{F}_\theta(x) - \mathcal{F}_\theta(x + \delta(x))\| \leq \epsilon,$$

for a small $\epsilon > 0$, indicating that the representation is stable under domain shift. Empirical evidence suggests that embeddings from foundation models occupy lower-dimensional manifolds with meaningful semantic structure, thereby enabling effective clustering, retrieval, and few-shot learning. Furthermore, pretraining on multi-task and multi-domain data implicitly encourages disentanglement of anatomical and pathological features, which is crucial for reliable medical AI. For instance, a model exposed to chest X-rays, brain MRIs, and abdominal CTs during pretraining can learn to isolate modality-agnostic patterns, such as lesion shape or tissue density, from modality-specific artifacts [46].

Challenges and Open Questions

Despite their promise, foundation models are not without limitations. Their performance in out-of-distribution (OOD) settings still depends on the diversity and quality of the pretraining corpus. Moreover, the high capacity and opacity of these models raise concerns about interpretability, fairness, and trustworthiness, especially in high-stakes medical contexts. There is also an ongoing debate about how best to measure generalization in foundation models: whether traditional domain generalization benchmarks suffice, or whether new evaluation frameworks are needed that reflect clinical deployment scenarios [47]. Additionally, the effectiveness of domain adaptation strategies on top of foundation models remains an open area of research. Some studies suggest that naive fine-tuning may degrade pretrained representations ("catastrophic forgetting"), while others advocate for adapter-based or contrastive re-alignment techniques. Understanding when and how to adapt foundation models without losing generality is critical for their widespread adoption in real-world healthcare systems [48]. In the next section, we turn to empirical studies and benchmarks that evaluate the domain robustness of foundation models across a variety of medical imaging tasks and institutions.

5 Benchmarking and Empirical Evaluation of Foundation Models for Domain Adaptation and Generalization

The rigorous evaluation of foundation models in the context of domain adaptation and generalization is essential for understanding their practical utility in medical imaging. Unlike traditional models, which are typically benchmarked on narrow tasks and single-domain datasets, foundation models are evaluated across a broad spectrum of tasks, modalities, institutions, and geographies. These empirical assessments aim to quantify the robustness, adaptability, and fairness of foundation models under diverse conditions, including both in-distribution (ID) and out-of-distribution (OOD) settings [49].

Evaluation Protocols and Metrics

Benchmarking foundation models requires carefully designed protocols that capture real-world variability. Common strategies include zero-shot evaluation, few-shot fine-tuning, and domain-specific adaptation, often under cross-institutional or cross-modality scenarios [50]. Let $\mathcal{D}^{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$ and $\mathcal{D}^{\text{test}} = \{(x_j, y_j)\}_{j=1}^{M}$ represent the training and evaluation datasets, respectively, where $\mathcal{D}^{\text{test}}$ may come from a different domain than $\mathcal{D}^{\text{train}}$ (i.e., $P^{\text{train}}(X, Y) \neq P^{\text{test}}(X, Y)$). Performance is typically measured using task-appropriate metrics, such as accuracy, AUC (Area Under the ROC Curve), Dice coefficient, Hausdorff distance, or mean Intersection-over-Union (mIoU). In domain generalization, the average performance across multiple target domains $\{\mathcal{D}^{(t)}\}_{t=1}^{T}$ is reported:

$$\text{Generalization Score} = \frac{1}{T}\sum_{t=1}^{T} \text{Metric}(f_\theta, \mathcal{D}^{(t)}),$$

where f_θ denotes the foundation model or its adapted version [51]. In domain adaptation settings, relative gains over source-only baselines or upper-bound models trained on target data are also analyzed. For example, if Acc^{SO} and Acc^{UB} denote source-only and upper-bound accuracies, the adaptation effectiveness Δ_{DA} can be defined as:

$$\Delta_{\text{DA}} = \frac{\text{Acc}^{\text{DA}} - \text{Acc}^{\text{SO}}}{\text{Acc}^{\text{UB}} - \text{Acc}^{\text{SO}}} [52].$$

Cross-Domain and Cross-Institutional Studies

Recent studies have systematically benchmarked foundation models across a range of domains and institutions. For example, multi-site datasets like MIDRC, MIMIC-CXR, CheXpert, and RSNA are used to evaluate performance across hospitals with different equipment, patient populations, and imaging protocols. These settings introduce domain shifts in terms of spatial resolution, radiographic quality, annotation style, and disease prevalence [53]. Let f_θ be a pretrained foundation model and let $\mathcal{D}^{\text{site}_k}$ denote the evaluation dataset from site k [54]. A generalization curve can be plotted as:

$$G(k) = \text{Metric}(f_\theta, \mathcal{D}^{\text{site}_k}),$$

which quantifies degradation as the domain distance between training and evaluation sites increases [55]. Empirical evidence shows that foundation models often outperform conventional supervised baselines on unseen sites without requiring domain-specific fine-tuning, indicating a strong degree of domain-agnostic representation learning [56].

Cross-Modality and Multi-Task Generalization

Another crucial axis of evaluation is cross-modality transfer [57]. Models pretrained on one modality (e.g., chest X-rays) are tested on another (e.g., CT scans or MRIs) to assess representational universality. Let $\mathcal{X}^{(m)}$ denote the image space corresponding to modality m, and suppose a model is pretrained on $\mathcal{X}^{(m_1)}$ and evaluated on $\mathcal{X}^{(m_2)}$ [58]. The transferability score $\mathcal{T}(m_1 \to m_2)$ is defined as:

$$\mathcal{T}(m_1 \to m_2) = \text{Metric}(f_\theta, \mathcal{D}^{(m_2)}) - \text{Metric}_{\text{scratch}}(\mathcal{D}^{(m_2)}),$$

where the second term denotes the performance of a model trained from scratch on the target modality [59]. Foundation models such as those trained with masked autoencoders (e.g., Med-MAE) or multi-modal contrastive learning (e.g., MedCLIP, GLoRIA) have demonstrated impressive cross-modality capabilities, sometimes matching or exceeding modality-specific models after light fine-tuning [60]. Additionally, foundation models are often benchmarked across multiple tasks, such as classification, segmentation, detection, and report generation. This multi-task evaluation serves to test the richness of the learned representations [61]. A multi-task generalization score can be computed as:

$$\text{MTGS} = \sum_{i=1}^{N_{\text{tasks}}} w_i \cdot \text{Metric}_i(f_\theta, \mathcal{D}_i),$$

where w_i is a task-specific weight, often normalized so that $\sum w_i = 1$. This score aggregates performance across tasks like lung disease classification, tumor segmentation, and anatomical landmark detection, providing a holistic measure of generalization.

Failure Modes and Robustness Stress Tests

Despite their success, foundation models are not immune to domain-specific vulnerabilities [62]. Empirical evaluations often include stress tests such as adversarial perturbations, outlier detection, and subpopulation bias analysis [63]. For instance, subgroup-specific metrics are computed across age, gender, ethnicity, or comorbidity strata to identify performance disparities:

$$\Delta_{\text{bias}} = \max_{g \in \mathcal{G}} \left| \text{Metric}_g(f_\theta) - \text{Metric}_{\text{avg}}(f_\theta) \right|,$$

where $\mathcal{G}$ denotes demographic groups and $\text{Metric}_{\text{avg}}$ is the global average [64]. Foundation models pretrained on diverse and balanced datasets tend to exhibit lower values of Δ_{bias}, though the degree of fairness is influenced by training objectives and data curation. Moreover, robustness under corruptions (e.g., Gaussian noise, motion blur, occlusion) is assessed using benchmarks like ImageNet-C adapted for medical imaging. Perturbed datasets $\tilde{\mathcal{D}}^{(c)}$ are constructed for each corruption type c, and the robustness degradation is calculated as:

$$\Delta_{\text{robust}} = \frac{1}{C} \sum_{c=1}^{C} \left(\text{Metric}(\mathcal{D}) - \text{Metric}(\tilde{\mathcal{D}}^{(c)}) \right),$$

where C is the number of corruption types [65]. Foundation models have been shown to maintain relatively stable performance under such stressors, especially when pretrained with augmentations or adversarial robustness objectives.

Limitations of Current Benchmarks

While empirical evaluations offer valuable insights, existing benchmarks still fall short of capturing the full complexity of clinical deployment. Most benchmarks assume clean and structured datasets, whereas real-world systems must handle missing data, ambiguous labels, and variable acquisition protocols [66]. Furthermore, many evaluations lack longitudinal and temporal variability, which is crucial for assessing generalization over time [67]. Another limitation is the narrow scope of some public datasets. For example, while CheXpert and MIMIC-CXR are widely used, they both originate from academic medical centers in the United States and may not generalize to low-resource settings or underrepresented populations. To address this, efforts are underway to construct federated and decentralized benchmarks that capture global heterogeneity. In the following section, we discuss emerging strategies that combine the strengths of classical domain adaptation with foundation model pretraining to further enhance generalization in medical imaging systems.

6 Hybrid Strategies: Combining Foundation Models with Classical Domain Adaptation Techniques

While foundation models offer powerful, general-purpose representations through large-scale pretraining, there remains substantial value in combining these with classical domain adaptation techniques. This hybrid approach can improve both the robustness and specificity of medical imaging models, especially in settings where domain shifts are complex, subtle, or underrepresented during pretraining. Classical domain adaptation (DA) methods—including feature alignment, domain-invariant learning, adversarial training, and discrepancy minimization—can be applied on top of foundation model embeddings to enhance cross-domain performance without retraining the entire model [68].

Feature-Space Alignment on Pretrained Embeddings

Let $\mathcal{F}_\theta : \mathcal{X} \to \mathbb{R}^d$ denote a frozen foundation model encoder, and let $\mathcal{Z}^{\text{src}} = \{\mathcal{F}_\theta(x_i^{\text{src}})\}_{i=1}^n$ and $\mathcal{Z}^{\text{tgt}} = \{\mathcal{F}_\theta(x_j^{\text{tgt}})\}_{j=1}^m$ be the sets of source and target domain embeddings, respectively [69]. Classical DA approaches attempt to align these distributions in feature space using domain-invariant transformations or alignment losses [70]. For instance, feature centering and whitening (ZCA), or more sophisticated alignment using Maximum Mean Discrepancy (MMD), can be applied:

$$\mathcal{L}_{\text{MMD}}(\mathcal{Z}^{\text{src}}, \mathcal{Z}^{\text{tgt}}) = \left\| \frac{1}{n} \sum_{i=1}^{n} \phi(z_i^{\text{src}}) - \frac{1}{m} \sum_{j=1}^{m} \phi(z_j^{\text{tgt}}) \right\|^2,$$

where ϕ is a feature map to a reproducing kernel Hilbert space (RKHS). This loss can be minimized by adapting a lightweight transformation layer or adapter $\mathcal{A}_\phi$ applied to the frozen features: $\hat{z} = \mathcal{A}_\phi(z)$. Alternatively, adversarial domain adaptation methods can be used to enforce domain confusion. A domain discriminator D_ψ is trained to distinguish between source and target embeddings, while the encoder or adapter is trained to fool it:

$$\min_\phi \max_\psi \mathbb{E}_{z^{\text{src}}}[\log D_\psi(\mathcal{A}_\phi(z^{\text{src}}))] + \mathbb{E}_{z^{\text{tgt}}}[\log(1 - D_\psi(\mathcal{A}_\phi(z^{\text{tgt}})))].$$

This adversarial loss complements the supervised or contrastive objectives used during foundation model training and improves feature invariance to domain shifts.

Adapters and Lightweight Re-parameterizations

One effective method for leveraging foundation models in domain adaptation is through adapters—parameter-efficient modules inserted within the model architecture that can be fine-tuned with minimal computation. Let $\mathcal{F}_\theta^{\text{frozen}}$ be a frozen backbone and Adapter_ϕ be a trainable function with significantly fewer parameters than θ. The output embedding becomes:

$$z = \text{Adapter}_\phi(\mathcal{F}_\theta^{\text{frozen}}(x)),$$

and the optimization proceeds only over ϕ. Popular adapter designs include bottleneck layers, residual low-rank projections (as in LoRA), and side-tuning. The domain-specific adapters can be stored separately, enabling modular reuse across different deployment environments without duplicating the entire model. Adapters are particularly well-suited to few-shot domain adaptation, where the target dataset $\mathcal{D}^{\text{tgt}}$ is small and labeled. The loss function then combines classification loss on $\mathcal{D}^{\text{tgt}}$ and a regularization term to prevent overfitting:

$$\mathcal{L}_{\text{total}} = \sum_{(x,y)\in\mathcal{D}^{\text{tgt}}} \ell(\text{Adapter}_\phi(\mathcal{F}_\theta(x)), y) + \lambda\|\phi\|^2.$$

Prompt Tuning for Visual-Language Foundation Models

In the context of vision-language foundation models, prompt tuning has emerged as a strategy for domain adaptation without altering the core model weights. A learnable prompt vector $p \in \mathbb{R}^k$ is prepended to the token sequence input into the image or text encoder. For instance, for an image x, the modified input becomes $[p; x]$, and the encoder $\mathcal{F}_\theta$ processes this augmented input. The prompt is optimized to adapt the model to the target domain:

$$\min_p \sum_{(x,y)\in\mathcal{D}^{\text{tgt}}} \ell(\mathcal{H}(\mathcal{F}_\theta([p; x])), y),$$

where $\mathcal{H}$ is the classification head. Prompt tuning is highly parameter-efficient and has been shown to achieve competitive results in zero- and few-shot settings with minimal domain-specific supervision.

Cross-Domain Contrastive Re-Alignment

Another promising strategy involves contrastive re-alignment of pretrained foundation embeddings across source and target domains. The goal is to refine representations by encouraging semantic consistency across domains, even in the absence of shared labels. Let $(x_i^{\text{src}}, x_j^{\text{tgt}})$ be a pair of semantically similar images from different domains. The re-alignment loss is:

$$\mathcal{L}_{\text{cross-contrast}} = -\log \frac{\exp(\text{sim}(z_i^{\text{src}}, z_j^{\text{tgt}})/\tau)}{\sum_k \exp(\text{sim}(z_i^{\text{src}}, z_k^{\text{tgt}})/\tau)},$$

where $z_i^{\text{src}} = \mathcal{F}_\theta(x_i^{\text{src}})$ and similarly for z_j^{tgt}. Positive pairs can be mined using pseudo-labeling, clustering, or domain-agnostic pretext tasks such as anatomical part matching [71]. These methods enhance the alignment of semantically equivalent representations while preserving the benefits of large-scale pretraining [72].

Self-Training and Pseudo-Labeling on Foundation Features

Self-training remains a robust and simple domain adaptation technique that can benefit from the high-quality features produced by foundation models [73]. In this setup, the pretrained model is first used to generate pseudo-labels $\hat{y}_j$ for unlabeled target domain examples x_j^{tgt}. The model is then fine-tuned on this pseudo-labeled dataset:

$$\mathcal{D}^{\text{pseudo}} = \{(x_j^{\text{tgt}}, \hat{y}_j) \mid \hat{y}_j = \arg\max f_\theta(x_j^{\text{tgt}})\}[74].$$

Given the strong calibration properties of foundation models, pseudo-labels tend to be more reliable than those generated by task-specific networks trained from scratch [75]. Confidence thresholding and ensembling can further improve label quality and reduce noise propagation [76].

Challenges and Considerations in Hybrid Methods

Despite their promise, hybrid strategies are not without challenges. One issue is the potential conflict between the learned inductive biases of the foundation model and the adaptation mechanism. Aggressive domain-specific fine-tuning or adversarial alignment may distort the pretrained embedding space, leading to catastrophic forgetting or loss of generality. To mitigate this, careful regularization and orthogonality constraints can be applied to the adaptation parameters. Another consideration is the computational and data efficiency of hybrid methods [77]. While many are designed to be lightweight, techniques like adversarial training or contrastive re-alignment can introduce significant complexity and instability.

Moreover, effective hybridization often requires labeled or at least paired data across domains, which may not always be available [78]. Finally, there is a need for standardized evaluation of hybrid strategies to determine when and how they improve over standalone foundation models [79]. This includes ablation studies, efficiency metrics, and domain robustness benchmarks that account for clinical variability and operational constraints. In the subsequent section, we will examine case studies and applications of foundation model adaptation in real-world clinical scenarios, including radiology, pathology, and multi-modal diagnostic pipelines.

7 Case Studies and Clinical Applications

The practical deployment of foundation models in clinical settings reveals their transformative potential, particularly when combined with domain adaptation strategies tailored to medical environments [80]. This section highlights case studies across multiple imaging modalities and clinical workflows, focusing on the strengths and limitations of these models in real-world contexts. We explore how foundation models, equipped with large-scale generalization and adaptability, are being used to enhance diagnostic accuracy, enable scalable annotation, support rare disease detection, and facilitate multi-modal decision-making [81].

Foundation Models in Chest Radiography

One of the most prominent use cases for vision-language foundation models is in chest radiography. Large-scale pretraining on datasets such as MIMIC-CXR, CheXpert, and PadChest has enabled models like GLoRIA, BioViL, and MedCLIP to generalize across tasks such as zero-shot disease classification, report generation, and abnormality localization. Consider a radiograph $x \in \mathcal{X}^{\text{CXR}}$ and a label space $\mathcal{Y} = \{\text{Atelectasis}, \text{Effusion}, \text{Cardiomegaly}, \ldots\}$ [82]. A vision-language foundation model embeds the image into a latent space $z_x = \mathcal{F}_\theta^{\text{img}}(x)$ and the disease concepts into text embeddings $z_y = \mathcal{F}_\theta^{\text{txt}}(y)$. Diagnosis is performed via cosine similarity or dot product:

$$\hat{y} = \arg\max_{y \in \mathcal{Y}} \frac{z_x^\top z_y}{\|z_x\|\|z_y\|} [83].$$

Studies show that these models, without explicit task-specific supervision, often achieve performance close to fully supervised baselines, especially in large-scale triage and abnormality detection settings. Furthermore, fine-tuning on small subsets of institution-specific data via adapters or prompt tuning leads to substantial improvements in specificity and calibration, making them suitable for deployment in new hospitals or regions.

Histopathology and High-Resolution Vision Tasks

Digital pathology poses unique challenges for machine learning due to the gigapixel scale of whole-slide images (WSIs) and the fine-grained nature of cellular structures. Foundation models adapted for high-resolution vision (e.g., through Vision Transformers with local-global attention or patch aggregation) have demonstrated strong capabilities in tissue classification, mitosis detection, and biomarker prediction. Let a WSI W be partitioned into N patches $\{x_i\}_{i=1}^N$. A patch-level encoder $\mathcal{F}_\theta$ produces embeddings $z_i = \mathcal{F}_\theta(x_i)$, and these are aggregated into a slide-level representation:

$$Z_W = \text{Aggregate}(\{z_i\}_{i=1}^N), \quad \text{e.g., via attention pooling or MIL (multiple instance learning).}$$

Domain adaptation techniques such as stain normalization, self-supervised fine-tuning on in-house data, or contrastive clustering are critical for generalization due to variability in staining protocols, scanners, and tissue preparation. Studies have reported success in adapting foundation models trained on public datasets (e.g., CAMELYON16, TCGA) to private clinical datasets using lightweight adapters or pseudo-label refinement.

Cross-Modality Clinical Integration

Medical diagnosis frequently involves integrating information from multiple modalities—e.g., CT scans, MRIs, lab results, and radiology reports. Foundation models that support multi-modal learning (e.g., CLIP-based or BERT-augmented architectures) can serve as centralized backbones in such pipelines. Let $\mathcal{X}^{\text{image}}$, $\mathcal{X}^{\text{text}}$, and $\mathcal{X}^{\text{omics}}$ denote input spaces for imaging, textual reports, and omics data [84]. A multi-modal foundation model can be expressed as:

$$f_\theta(x^{\text{image}}, x^{\text{text}}, x^{\text{omics}}) = \mathcal{H}\left(\mathcal{F}_\theta^{\text{img}}(x^{\text{image}}), \mathcal{F}_\theta^{\text{txt}}(x^{\text{text}}), \mathcal{F}_\theta^{\text{omics}}(x^{\text{omics}})\right),$$

where $\mathcal{H}$ is a fusion head, often implemented via attention mechanisms or graph-based learning. Applications include cancer subtype classification, clinical decision support for rare diseases, and longitudinal patient modeling [85]. Case studies in oncology show that foundation models pretrained on one modality (e.g., radiology images) can be effectively adapted to integrate genomic profiles or pathology slides through modality-specific adapters. This capability supports personalized treatment planning and prognosis estimation, especially in resource-constrained environments [86].

Low-Resource and Global Health Settings

In low-resource settings, where data scarcity, label noise, and infrastructure limitations are common, foundation models offer a unique advantage due to their ability to generalize from minimal supervision. For example, zero-shot tuberculosis (TB) screening using foundation models trained on CXR images from high-resource settings has been piloted in sub-Saharan Africa and Southeast Asia [87]. Adaptation strategies in these environments often include self-training, weak supervision using noisy labels, and unsupervised domain alignment. An example workflow is:

1. Use a vision-language foundation model to produce soft labels or text-guided similarity scores on unlabeled CXR data.
2. Filter predictions with uncertainty or entropy thresholds.
3. Fine-tune a lightweight classifier on the pseudo-labeled data using local computing resources [88].

Such workflows have shown strong improvements in screening specificity and sensitivity without requiring high-bandwidth annotation pipelines. Furthermore, local clinicians can provide low-shot corrections that are used to fine-tune adapters or prompts, effectively tailoring the model to regional disease phenotypes and equipment.

Clinical Deployment and Regulatory Considerations

Despite promising results, deploying foundation models in clinical practice involves challenges beyond performance metrics [89]. Interpretability, robustness, regulatory compliance, and human-AI interaction must all be considered [90]. Many foundation models are being adapted with explainability modules (e.g., saliency maps, attention rollouts) and uncertainty quantification tools to meet clinical interpretability standards. Moreover, fine-tuning or adapting foundation models on institution-specific data must comply with data governance, privacy regulations (e.g., HIPAA, GDPR), and model auditability requirements. Hospitals increasingly demand models whose adaptation pathways are transparent and whose decision-making can be traced, versioned, and re-certified as new data become available. In the next section, we turn to these considerations, outlining the ethical, regulatory, and operational challenges of deploying foundation models that have undergone domain adaptation in sensitive clinical workflows [91].

8 Ethical, Regulatory, and Operational Challenges

The deployment of foundation models adapted for domain-specific medical imaging tasks presents not only technical and clinical opportunities, but also substantial ethical, regulatory, and operational concerns. These challenges stem from the complex interplay of data privacy, model transparency, regulatory oversight, and the high stakes associated with medical decision-making. As models move from development environments into real-world clinical workflows, it becomes imperative to systematically address these multifaceted risks to ensure responsible AI deployment in healthcare [92].

Privacy and Data Governance in Domain Adaptation

Domain adaptation typically involves fine-tuning or calibrating models on data from a new clinical domain, which can introduce sensitive patient information into the training pipeline [93]. Even when only embeddings or low-dimensional feature representations are used, the potential for re-identification or information leakage persists [94]. Given that foundation models are often pretrained on large public datasets and subsequently adapted to private clinical data, robust governance frameworks must be implemented. Let $\mathcal{D}^{\text{priv}} = \{(x_i, y_i)\}$ denote a dataset from a protected medical domain [95]. Suppose a foundation model $\mathcal{F}_\theta$ is adapted on $\mathcal{D}^{\text{priv}}$ via parameter updates $\Delta\theta$. One must ensure that the resulting model $f_{\theta+\Delta\theta}$ does not inadvertently memorize or expose identifiable patient features [96]. Differential privacy (DP) techniques can be used to constrain updates:

$$\mathbb{P}[f_{\theta+\Delta\theta} \in \mathcal{S}] \leq e^{\varepsilon}\mathbb{P}[f_{\theta+\Delta\theta'} \in \mathcal{S}] + \delta,$$

for any neighboring datasets differing in a single record. However, integrating DP with high-capacity foundation models remains a nontrivial technical challenge due to gradient noise amplification and performance degradation, especially in few-shot adaptation regimes [97].

Explainability and Trust in Clinical Decision-Making

A recurring concern with deep foundation models—especially those adapted across domains—is their opacity. While large-scale pretraining imparts strong generalization, it also introduces layers of abstraction that complicate interpretability [98]. In clinical imaging, where decisions impact patient safety, the ability to trace model behavior and provide human-understandable justifications is crucial [99]. Let $f_\theta(x) = y$ be a prediction from an adapted model. The model's internal attention scores or gradient-based saliency maps $\mathcal{S}(x)$ are often used to generate visual explanations. However, domain shifts can disrupt these interpretations: saliency maps may highlight irrelevant or misleading regions in the target domain, especially if the adaptation is insufficient. Furthermore, prompt-tuned or adapter-based models may alter internal representations in unpredictable ways. To address this, post-hoc explainability frameworks (e.g., LIME, SHAP, Grad-CAM++) must be re-evaluated in the adapted domain, and local clinicians should be involved in validating the consistency and utility of the model's explanations. Moreover, research is emerging around inherently interpretable foundation architectures—such as modular reasoning networks and attention sparsification—that could enhance trustworthiness without sacrificing performance [100].

Regulatory Compliance and Certification

From a regulatory perspective, the use of domain-adapted foundation models introduces novel challenges. Existing frameworks such as the FDA's Software as a Medical Device (SaMD) or the European CE marking guidelines are primarily designed for static models with fixed decision boundaries [101]. Foundation models, by contrast, often evolve over time through continual adaptation, user interaction, or federated learning across institutions. Let f_θ^{v0} be the original certified model. Upon domain-specific adaptation to new data $\mathcal{D}^{\mathrm{site}}$, the updated model $f_{\theta+\Delta\theta}^{\mathrm{site}}$ may no longer fall under the original regulatory approval, requiring re-evaluation [102]. However, traditional regulatory processes are too slow and cumbersome to keep up with real-time model evolution. To reconcile adaptability with compliance, researchers and policymakers are exploring modular certification, where only specific components (e.g., adapters, prompts) are subjected to review. The certification process can then be scoped over subspaces of the model, denoted $\mathcal{M}_{\mathrm{cert}} \subset \theta$, such that:

$$f_\theta = f_{\mathcal{M}_{\mathrm{frozen}}} + f_{\mathcal{M}_{\mathrm{cert}}},$$

where $\mathcal{M}_{\mathrm{frozen}}$ remains unchanged and $\mathcal{M}_{\mathrm{cert}}$ is auditable. This approach can enable faster iteration and domain personalization while maintaining safety and transparency standards [103].

Bias, Fairness, and Clinical Equity

Adaptation processes may unintentionally amplify existing biases in both foundation models and target domain datasets [104]. If the pretraining corpus underrepresents certain demographics (e.g., pediatric populations, under-resourced regions), the model may exhibit systematically worse performance for those groups even after adaptation [105]. Additionally, localized adaptation may cause overfitting to majority phenotypes within the new domain, further marginalizing outliers or minority subgroups [106]. Let $x^{(g)}$ be a sample from demographic group g, and suppose the adapted model $f_{\theta+\Delta\theta}$ has group-wise error rates $\epsilon^{(g)}$. Fairness constraints seek to enforce:

$$|\epsilon^{(g_1)} - \epsilon^{(g_2)}| < \delta, \quad \forall g_1, g_2 \in \mathcal{G},$$

for a small tolerance δ. In practice, this can be achieved through adversarial debiasing, group re-weighting, or fairness-aware loss functions during adaptation. However, this requires demographic labels, which are often unavailable or inconsistently recorded in medical datasets. Bias audits, stratified performance evaluations, and clinician-in-the-loop assessments are essential to prevent inadvertent harm. Foundation models, due to their scale and opacity, must be subjected to stricter scrutiny than task-specific models, especially when used in high-risk scenarios such as cancer staging or emergency triage [107].

Operational Integration and Human-AI Collaboration

Finally, the successful use of adapted foundation models in clinical practice depends heavily on their integration into existing workflows and the dynamics of human-AI collaboration [108]. Even a highly accurate model can fail operationally if it does not align with clinician expectations, EMR systems, or radiology PACS environments. Operational deployment involves challenges such as latency, compatibility, real-time inference, model update pipelines, and feedback capture [109]. Continuous learning from clinician feedback poses another complexity: how to incorporate corrections, overrides, or annotations into the model without breaking compliance or introducing drift. One approach is to maintain a shadow model f_θ^{shadow} that learns passively from feedback and is periodically benchmarked before promotion to production. This allows for safe experimentation and evaluation without affecting patient care. Moreover, adaptive user interfaces, which explain the model's reasoning and uncertainty in context-specific terms, are key to fostering trust and effective decision support [110].

Summary and Path Forward

Ethical, regulatory, and operational challenges represent the primary barriers to the widespread adoption of domain-adapted foundation models in medical imaging. These challenges cannot be resolved by technical solutions alone; rather, they require interdisciplinary collaboration among clinicians, engineers, ethicists, and regulators [111]. As foundation models continue to evolve, proactive governance, transparent adaptation protocols, and participatory design will be essential to ensure that their benefits are realized safely, equitably, and at scale. In the following section, we present a forward-looking discussion on open research directions and future opportunities at the intersection of foundation models, domain adaptation, and generalization in medical imaging [112].

9 Open Research Directions and Future Outlook

As foundation models continue to reshape the landscape of medical imaging, their potential to revolutionize diagnostic workflows, clinical decision-making, and personalized medicine becomes increasingly clear. However, fully realizing this promise necessitates solving a constellation of open research problems that span theory, methodology, systems design, and translational science [113]. This section surveys key frontiers in domain adaptation and generalization for foundation models in medical imaging, laying out directions for future exploration [114].

Theoretical Foundations of Domain Generalization

Despite the empirical success of foundation models in cross-domain generalization, the theoretical underpinnings of why and when these models generalize remain underdeveloped. Existing theories in domain adaptation—such as the $\mathcal{H}$-divergence and discrepancy bounds—struggle to scale to high-dimensional, overparameterized models with billions of parameters. Let $\mathcal{D}_S$ and $\mathcal{D}_T$ represent the source and target domains respectively, and let $\mathcal{H}$ be a hypothesis class [115]. Traditional generalization bounds in domain adaptation rely on quantities such as:

$$\epsilon_T(h) \leq \epsilon_S(h) + d_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) + \lambda,$$

where $d_{\mathcal{H}}(\cdot, \cdot)$ is a divergence between domains and λ captures the error of the optimal shared hypothesis. In foundation models, however, h is often instantiated as a deep encoder-decoder stack with attention layers, trained over diverse tasks and domains [116]. How to appropriately define $d_{\mathcal{H}}$ in this setting, and whether such bounds are meaningful, is still an open question. Recent efforts have explored generalization in terms of representation alignment, mutual information, and contrastive invariance, but a unified theory applicable to foundation models across clinical domains is lacking. Closing this gap would provide stronger guarantees for safety-critical applications and guide the design of future architectures.

Efficient and Safe Adaptation Protocols

Adapting foundation models to new medical domains must balance performance, efficiency, and safety. Full fine-tuning is often computationally infeasible and risks overfitting, while lightweight methods—such as adapter modules, LoRA (Low-Rank Adaptation), or prompt tuning—may not always achieve the desired task fidelity. Let $\mathcal{F}_\theta$ be a foundation model and ϕ be the parameters introduced for adaptation. A central challenge is finding optimal configurations of ϕ that maximize target-domain performance under budget constraints:

$$\min_{\phi} \mathbb{E}_{(x,y)\sim\mathcal{D}_T} \mathcal{L}(f_{\theta,\phi}(x), y), \quad \text{subject to} \quad \text{Size}(\phi) \leq \kappa, \quad \text{Time}(\phi) \leq \tau,$$

where κ and τ are resource budgets (e.g., memory, latency). Additionally, how to validate these protocols when access to labeled target data is limited remains an unsolved problem. Few-shot cross-validation, synthetic supervision, or clinician-in-the-loop bootstrapping are promising but require further refinement. The development of universal, plug-and-play adaptation toolkits for clinical institutions—similar to AutoML but tailored for medical workflows—remains a critical area for future work.

Benchmarking Across Clinical Domains and Tasks

While foundation models have been evaluated on individual tasks and datasets, comprehensive, community-curated benchmarks that span multiple imaging modalities, disease types, and geographical regions are still rare. Such benchmarks are essential for fair and reproducible comparisons of domain adaptation strategies and for identifying where foundation models fall short [117]. A desirable benchmark suite would contain:

- A diverse set of domains $\{\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_n\}$ including different imaging devices, patient populations, and clinical workflows [118].
- Standardized adaptation settings: zero-shot, few-shot, and full supervision [119].
- Multi-dimensional metrics covering accuracy, calibration, fairness, interpretability, and computational cost [120].
- Evaluation protocols for continual learning and robustness under domain drift.

Efforts such as MedFMC, MedDG, and CXR-FM are early steps in this direction, but there is significant room for expansion [121]. Open-source collaborations between academia, hospitals, and industry could accelerate progress and help identify systemic blind spots in current model capabilities [122].

Cross-Modal and Multi-Omic Adaptation

A major opportunity lies in extending foundation models to handle richer forms of medical data beyond imaging [104]. Genomics, histopathology, electronic health records (EHRs), and wearable sensor streams each offer complementary perspectives on patient health. Developing unified models that can ingest and generalize across these modalities requires advances in data fusion, representation alignment, and task disentanglement [123]. Let inputs be drawn from modalities $\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_m$ with aligned patient indices [124]. A foundation model must learn modality-specific encoders $\mathcal{F}_i : \mathcal{X}_i \to \mathbb{R}^{d_i}$ and a shared fusion mechanism $\mathcal{G}(\cdot)$ such that:

$$f(x_1, \ldots, x_m) = \mathcal{G}(\mathcal{F}_1(x_1), \ldots, \mathcal{F}_m(x_m)).$$

Learning such models in the presence of missing modalities, noise, and institutional biases remains an open challenge. Moreover, how to adapt such models when only a subset of modalities is available in the target domain (e.g., imaging but not genomics) introduces novel semi-supervised and imputation-based techniques into the domain adaptation landscape.

Human-Centered Design and Participatory Adaptation

A foundational principle for future work must be the active involvement of end-users—clinicians, radiologists, and healthcare providers—in the design and adaptation process. Participatory machine learning frameworks, where users can iteratively refine, evaluate, and critique models during deployment, hold promise for aligning model behavior with clinical expectations [125]. This includes:

- Visualization tools that surface model uncertainty, feature importance, and training provenance [126].
- Feedback mechanisms that allow clinicians to flag mispredictions or suggest corrections.
- Interfaces that support explainable counterfactuals or "what-if" analyses to test clinical hypotheses.

The development of such systems requires interdisciplinary collaboration across HCI, machine learning, clinical science, and regulatory policy. As the field matures, the shift from purely model-centric research to human-centered adaptation protocols will be critical to achieving safe, equitable, and trusted AI in medicine.

Conclusion

Foundation models have unlocked unprecedented capabilities in medical imaging, enabling flexible, scalable, and increasingly generalizable solutions across diverse clinical contexts. Yet, significant challenges remain in ensuring that these models can be adapted responsibly, robustly, and equitably to new domains [127]. Future research must advance both the science of domain adaptation and the infrastructure for its deployment, building models that not only perform well, but also serve patients, clinicians, and society at large. In closing, we advocate for a new paradigm in medical AI—one where foundation models are not static artifacts, but dynamic collaborators, continuously learning and adapting under the guidance of rigorous science and human insight.

10 Conclusion

The integration of foundation models into the medical imaging ecosystem marks a profound shift in how artificial intelligence is designed, developed, and deployed for clinical tasks. These models, pre-trained on massive corpora of visual and multimodal data, offer a unique opportunity to transcend traditional limitations of narrow, task-specific solutions. Through domain adaptation and generalization techniques, foundation models can be tailored to diverse healthcare settings, imaging modalities, and patient populations, promising significant improvements in diagnostic performance, scalability, and clinical reach.

However, the journey from general-purpose pretraining to clinically viable adaptation is complex and fraught with theoretical, methodological, and operational challenges. Throughout this survey, we have examined the foundational principles of domain adaptation—ranging from statistical alignment to representation learning—and their adaptation to the context of high-capacity foundation models. We have explored practical strategies, including prompt tuning, adapter modules, few-shot fine-tuning, and continual learning, as well as their respective trade-offs in performance, data efficiency, and interpretability.

We have also critically reviewed current benchmarks and empirical evaluations, highlighting the limitations of existing datasets and the need for standardized protocols that reflect real-world clinical variation. Our survey emphasized the importance of addressing ethical and regulatory concerns, particularly in ensuring patient privacy, mitigating bias, and enabling safe deployment. Moreover, we have identified promising directions for future research, including theoretical advancements in generalization theory for overparameterized models, scalable and modular adaptation protocols, multimodal integration, and human-centered design paradigms.

As foundation models become increasingly central to medical AI pipelines, the field must shift toward a principled, multidisciplinary approach to adaptation and generalization—one that integrates advances in machine learning with deep clinical knowledge, rigorous evaluation, and participatory governance. This will require not only technical innovation but also new norms for collaboration, reproducibility, and accountability.

In conclusion, domain adaptation and generalization are not merely auxiliary steps in the deployment of foundation models—they are critical levers for ensuring that these models can meaningfully support clinical decision-making across diverse and evolving healthcare environments. By embracing this challenge, the medical AI community can help build a future where intelligent systems are not only powerful but also inclusive, safe, and aligned with the highest standards of patient care.

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